

Lecture Notes III

Course: General Topology

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Topic: Connectedness

Proof 4.1 (Characterisation of connected spaces). Suppose X is connected and that U is open and closed. Then $V = X \setminus U$ is also open and closed and if U and V were both nonempty, they would disconnect X , a contradiction, so either U is empty or V is empty, meaning that either $U = X$ or $U = \emptyset$. Conversely, suppose that X is disconnected, then we can write $X = U \amalg V$ for some U, V open, nonempty and disjoint, this means that the complement of U is V and the complement of V is U so both are also closed and neither of them is $\emptyset$ so none of them can be X .

Proof 4.2 (Continuous maps from connected spaces to discrete ones must be constant). Let Y be a discrete space and let X be connected, then for $f : X \rightarrow Y$ continuous we have that for any $x \in X$ we can choose $c = f(x)$ and thus the preimage $f^{-1}(\{c\})$ must be open and closed in X , because $\{c\}$ is both open and closed in Y , but this means that either $f^{-1}(\{c\})$ is the whole space X or is empty, but it cannot be empty because $x \in f^{-1}(\{c\})$ so it is X and thus the map is constant.

Proof 4.3 (Main theorem on connectedness). Let $f : X \rightarrow Y$ be a continuous map and consider it as a function $f : X \rightarrow f(X)$ so that f is surjective, then if $f(X)$ is disconnected, it is the disjoint union of two disjoint nonempty open subsets U, V , then by continuity of f , $f^{-1}(U)$ and $f^{-1}(V)$ are open, disjoint and nonempty so they disconnect X .

Proof 4.4 (Connectedness is a topological property). Let X and Y be homeomorphic, then use the previous result and the homeomorphism φ from X to Y which by surjectivity, $\varphi(X) = Y$.

Proof 4.5 (Properties of connected spaces). For this, recall that a subset of X is said to be connected if it is connected as a subspace.

i) Let A be a connected subset of X and let U, V be disjoint open subsets of X , then if A contained a point of U and a point in V , then $A \cap U$ and $A \cap V$ would disconnect A , a contradiction, thus A has either no points of U or no points of V .

ii) Let A be a dense subset of X and suppose that U, V disconnect X . Then by i) we have that A is contained in either U or V , wlog suppose that $A \subseteq U$, then $X = \overline{A} \subseteq \overline{U}$, but since the complement of U is V , then U is also closed so that $\overline{U} = U$, so $X \subseteq U$ but also $U \subseteq X$ so $X = U$ and therefore V must be empty, a contradiction.

iii) Suppose A is a connected subset of X and $A \subseteq B \subseteq \overline{A}$. Since A is dense in $\overline{A}$, A is dense in B so that B is connected by ii), the remaining result follows now from taking $B = \overline{A}$.

iv) Suppose p is the point in common and suppose that U, V are disjoint open subsets of $\bigcup_{\alpha \in I} B_{\alpha}$, suppose wlog that $p \in U$, which by i) yields that $\bigcup_{\alpha \in I} B_{\alpha}$ is contained in U , meaning that V would need to be empty and therefore there cannot be a disconnection of $\bigcup_{\alpha \in I} B_{\alpha}$.

v) Let X, Y be connected and suppose that U, V disconnect $X \times Y$, then for any point $(x_0, y_0) \in U$ we have that $\{x_0\} \times Y$ is homeomorphic to Y ($\varphi(x_0, y) = y$) so $\{x_0\} \times Y$ is connected and contains the point (x_0, y_0) so $\{x_0\} \times Y \subseteq U$ by i). Next, for each $y \in Y$, the set $X \times \{y\}$ is connected by a similar argument and contains (x_0, y) for any y by the previous remark so $X \times \{y\} \subseteq U$ for each $y \in Y$, and since

$$X \times Y = \bigcup_{y \in Y} X \times \{y\} \subseteq U$$

we conclude that $U = X \times Y$ and thus V is empty, a contradiction.

vi) If X is a topological space, then the quotient space $X/\sim$ is defined by the map $q : X \rightarrow X/\sim$ which is surjective and continuous by definition of the induced topology, thus $q(X) = X/\sim$ is connected by Theorem 4.3.

Proof 4.6 (Connectedness on $\mathbb{R}$). Let $J \subseteq \mathbb{R}$, then if J contains only one point (singleton) then $J = \{j\}$ cannot be written as the disjoint union of two nonempty open sets so it is connected. If J

contains at least two points then it is either an interval or not, first assume that J is an interval and that it is disconnected, then there are $U, V \subseteq \mathbb{R}$ open in $\mathbb{R}$ such that $U \cap J$ and $V \cap J$ disconnect J , choose $a \in U \cap J$ and $b \in V \cap J$, then assume $a < b$, then $[a, b] \subseteq J$ because J is an interval (meaning that any point $a < c < b$ is also in J), but since U, V are open, there exists $\varepsilon > 0$ such that $[a, a + \varepsilon) \subseteq U$ and $(b - \varepsilon, b] \subseteq V$ so let $c = \sup(U \cap [a, b])$ but $[a, a + \varepsilon) \cap [a, b] \subseteq U \cap [a, b]$ with $[a, a + \varepsilon) \cap [a, b] = [a, \min\{a + \varepsilon, b\}]$ but since U and V are disjoint (relative to J), the choice of ε implies that $\min\{a + \varepsilon, b\} = a + \varepsilon$ so we get $[a, a + \varepsilon) \subseteq U \cap [a, b]$ and thus $a + \varepsilon \leq c$, similarly, one gets that $c \leq b - \varepsilon$ so $a + \varepsilon \leq c \leq b - \varepsilon$ so c is in $[a, b] \subseteq J$ and thus $c \in U \cup V$ so that either $c \in U$ or $c \in V$ but not both because U, V are disjoint, if $c \in U$ then there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq U$ so $c + \delta \in U$ and because U and V are disjoint, also $c + \delta < b$ so $c + \delta \in [a, b]$ meaning that c is not an upper bound for $U \cap [a, b]$ because $c < c + \delta$, a contradiction, and similarly, if $c \in V$, then $(c - \delta', c + \delta') \subseteq V$ for some δ' so $c - \delta'$ would be a smaller upper bound for $U \cap [a, b]$, again, a contradiction. We thus conclude that J must be connected.

Conversely, if J is not an interval then there exists $a < c < b$ with $a, b \in J$ and $c \notin J$ so that $(-\infty, c) \cap J$ and $(c, \infty) \cap J$ disconnect J (they are open in J) and the proof is done.

Proof 4.7 (Generalised Bolzano's Intermediate Value Theorem). Apply the main theorem on connectedness and the previous result to get that $f(X)$ must be a singleton or an interval, if it is a singleton then $f(p) = f(q)$ for any $p, q \in X$ so that the function is constant and the theorem holds, if it is an interval then given any $p, q \in X$ we have $f(p), f(q) \in f(X)$ so that for any other m with $f(p) < m < f(q)$ or $f(q) < m < f(p)$ we must have $m \in f(X)$ because $f(X)$ is an interval.

Proof (Lemma). $(\Rightarrow)$ Suppose there is a map $f : [0, 1] \rightarrow X$ from p to q for any $p, q \in X$, then define $\phi : [a, b] \rightarrow [0, 1]$ by

$$\phi(t) = \frac{t - a}{b - a}$$

so that $g = f \circ \phi$ is the required map. $(\Leftarrow)$ Conversely, suppose that there is a map $g : [a, b] \rightarrow X$ such that $g(a) = p$ and $g(b) = q$, then define $\sigma : [0, 1] \rightarrow [a, b]$ by

$$\sigma(t) = (b - a)t + a$$

so that $f = g \circ \sigma$ is the required map. Continuity in both cases follows from the fact that ϕ, σ are continuous and composition preserves continuity.

Proof 4.8 (Properties of path-connected spaces). i) Let $g : X \rightarrow Y$ be a continuous map and let X be path-connected, then for any $y_1, y_2 \in g(X)$ we take $x_1 \in g^{-1}(\{y_1\}), x_2 \in g^{-1}(\{y_2\})$ so that $x_1, x_2 \in X$ and thus there exists a continuous map $f : I \rightarrow X$ such that $f(0) = x_1, f(1) = x_2$ and thus $g \circ f : I \rightarrow g(X)$ is a continuous map such that $(g \circ f)(0) = y_1$ and $(g \circ f)(1) = y_2$ and since y_1, y_2 were arbitrary, we conclude that $g(X)$ is path-connected.

ii) Let p be the point in common and thus for any $b_1, b_2 \in \bigcup_{\alpha \in I} B_\alpha$ we have that $b_1 \in B_\delta$ and $b_2 \in B_\gamma$ for some $\delta, \gamma \in I$, then since $p \in B_\alpha$ for each $\alpha \in I$, in particular, $p \in B_\delta \cap B_\gamma$ so by the assumption that each B is path-connected, take $f_\delta : [0, 1/2] \rightarrow B_\delta$ be a map from b_1 to p and $f_\gamma : [1/2, 1] \rightarrow B_\gamma$ be a map from p to b_2 , so that $f : [0, 1] \rightarrow \bigcup_{\alpha \in I} B_\alpha$

$$f(t) = \begin{cases} f_\delta(t) & t \in [0, 1/2], \\ f_\gamma(t) & t \in [1/2, 1], \end{cases}$$

is a path from b_1 to b_2 and the proof is complete.

iii) Let X, Y be path-connected spaces, then for any $p, q \in X \times Y$ define $f : [0, 1] \rightarrow X \times Y$ by

$$f = (f_X, f_Y)$$

with $f_X : [0, 1] \rightarrow X$ a map from $\pi^1(p)$ to $\pi^1(q)$ and $f_Y : [0, 1] \rightarrow Y$ a map from $\pi^2(p)$ to $\pi^2(q)$ so that f is clearly a map from p to q which is continuous by the characteristic property of the product topology, and thus $X \times Y$ is path-connected, as desired.

Proof 4.9 (Path-connectedness implies connectedness). Suppose X is path-connected and fix $p \in X$, then for any $q \in X$ take a map $f_q : I \rightarrow X$ from p to q so by Theorem 4.3 (Main Theorem on

Connectedness) we have that $f_q(I)$ is connected, but notice how $\bigcup_{q \in X} f_q(I) = X$ so that by Theorem 4.5(iv) we get that X is connected.

Proof 4.12 (Components form a partition). Let U, V be two distinct components of X and suppose that they are not disjoint, meaning that they share a point p , but by Theorem 4.5(iv) we have that $U \cup V$ is connected, but since $U \subseteq U \cup V$ and U is a component of X , then $U \not\subseteq U \cup V$ so $U \cup V = U$ and similarly, $V = U \cup V$ so $U = V$, a contradiction. We therefore conclude that either U, V are disjoint or they are equal, for cover, let $x \in X$, then there is at least one nonempty connected subset, namely $\{x\}$, so define the (nonempty) collection $\mathcal{A}_x$ to be the connected subsets of X that contain x , then their union $C_x \equiv \bigcup \mathcal{A}_x$ is connected because all the elements share x and it certainly is maximal, so it is a component and thus

$$X = \coprod_{x \in X} C_x$$

meaning that the components form a partition.

Proof 4.13 (Connectedness is equivalent to a single component). If X is connected, then we know that X is certainly maximal and nonempty so X is a component, but since the components form a partition, any other maximal connected subset would need to be empty, hence not a component so X is the only component. Conversely, if X has only one component, say C , then by the partition property $C = X$ but by definition, the components are connected so X is connected.

Proof 4.15 (Properties of components). i) Let $A \subseteq X$ be a component, then $\overline{A}$ is also connected and $A \subseteq A \subseteq \overline{A}$ so that by maximality we have $A = \overline{A}$ which is equivalent to A being closed.

ii) Let $A \subseteq X$ be a nonempty connected subset. Then since the components cover X by the previous theorem, A has a point in common with some component B and thus $A \cup B$ is connected by Theorem 4.5(iv) but $B \subseteq A \cup B$ so by maximality of the component B we conclude that $A \cup B = B$ which is equivalent to $A \subseteq B$ and the proof is done.

Proof 4.16 (Properties of path components). i) Suppose U, V are path components of X that are not disjoint, then $U \cup V$ is path-connected because U, V share a point in common, and thus $U = U \cup V = V$ by maximality (now on path-connectedness) so that V and U are either equal or disjoint. For the cover let $\mathcal{C}$ denote the collection of path components, then we claim that

$$\bigcup \mathcal{C} = X,$$

to prove such claim we note that clearly $\bigcup \mathcal{C} \subseteq X$, conversely, given any $x \in X$, we have that at least $\{x\}$ is path-connected (use Id) and thus define C_x to be the (nonempty by the previous remark) collection of all path-connected sets containing x , so that by Theorem 4.8(ii) we have that their union is path-connected (the point in common being x) and it certainly is maximal, the desired result $X \subseteq \bigcup \mathcal{C}$ now follows from the arbitrariness of x .

ii) Let Γ be a path component of X and thus Γ is nonempty and connected, thus by Theorem 4.15(ii) Γ is contained in a single component. For the other statement, let Λ be a component of X , then we claim that

$$\Lambda = \bigcup \mathcal{C}^\Lambda,$$

where $\mathcal{C}^\Lambda$ denotes the collection of path components of X contained in Λ . To prove this we, again, notice that clearly $\bigcup \mathcal{C}^\Lambda \subseteq \Lambda$, conversely, take $\lambda \in \Lambda$, then in particular $\lambda \in X$, so it must be in some path component P but hence this path component shares a point with Λ so that P is connected and hence $P \cup \Lambda$ is connected but by maximality of Λ we get $\Lambda = \Lambda \cup P$ which is equivalent to $P \subseteq \Lambda$ and thus $P \in \mathcal{C}^\Lambda$ so that any element of Λ is in some element of $\mathcal{C}^\Lambda$, completing the proof.

iii) Suppose A is nonempty a path-connected subset of X , and thus it contains points of a path component U (this is always true because the path components form a cover for X), then $U \cup A$ is path-connected and by maximality we have $U = U \cup A$ so $A \subseteq U$ and since path components are disjoint, A can only be contained in U .

Proof 4.17 (Manifolds are locally connected). Let M be a manifold. Given any $p \in M$ we know that there is a neighbourhood U homeomorphic (via, say, φ) to an open subset of $\mathbb{R}^n$, then take any open

neighbourhood V of p and thus $U \cap V$ is still an open neighbourhood of p so $\varphi(U \cap V)$ is open in $\mathbb{R}^n$ and contains $\varphi(p)$ so there exists $r > 0$ such that the (path-connected) ball $B_r(\varphi(p))$ is contained in $\varphi(U \cap V)$ and thus $N = \varphi^{-1}(B_r(\varphi(p)))$ is a path-connected subset which is open in $U \cap V$ but $U \cap V$ is open in M so by **Proposition 3.2(i)** N is an open neighbourhood that is path-connected (and thus connected) so that M is locally path-connected and thus locally connected.

Proof 4.18 (Properties of locally connected spaces). i) Let U be an open subset of X and $\mathcal{B}$ a basis thereof consisting of connected open subsets, then any open A in U is open in X and thus A is the union of some collection in $\mathcal{B}$ but the elements in the collection must be contained in U because A is, hence the subcollection $\mathcal{B}_U \subseteq \mathcal{B}$ of $\mathcal{B}$ of elements contained in U is a basis for U so U is locally connected.

ii) Let A be a component of X . If $p \in A$, then p has a connected neighbourhood U by local connectedness ($p \in X$ in particular) and thus $U \subseteq A$ by **Theorem 4.15(ii)**. We therefore conclude that each point of A has a neighbourhood contained in A , so A is open by **Theorem 2.7(vii)** and the proof is complete.

Proof 4.19 (Properties of locally path-connected spaces). i) Straightforward

ii), iii) same as in the previous theorem

iv) Let $p \in X$ and let A and B be the component and path component containing p , respectively. **Theorem 4.17(ii)** says that each path component is contained in a single component and since A, B share p , then $B \subseteq A$, and also says that A can be written as a disjoint union of path components, each of which is open in X by iii) in this theorem and thus open in A . If B is not the only path component in A , then the sets B and $A \setminus B$ disconnect A , a contradiction, so $A = B$.

v) X being connected is equivalent to it having only one component, which by iv) is equivalent to having only one path component, and this is in turn equivalent to it being path-connected.